

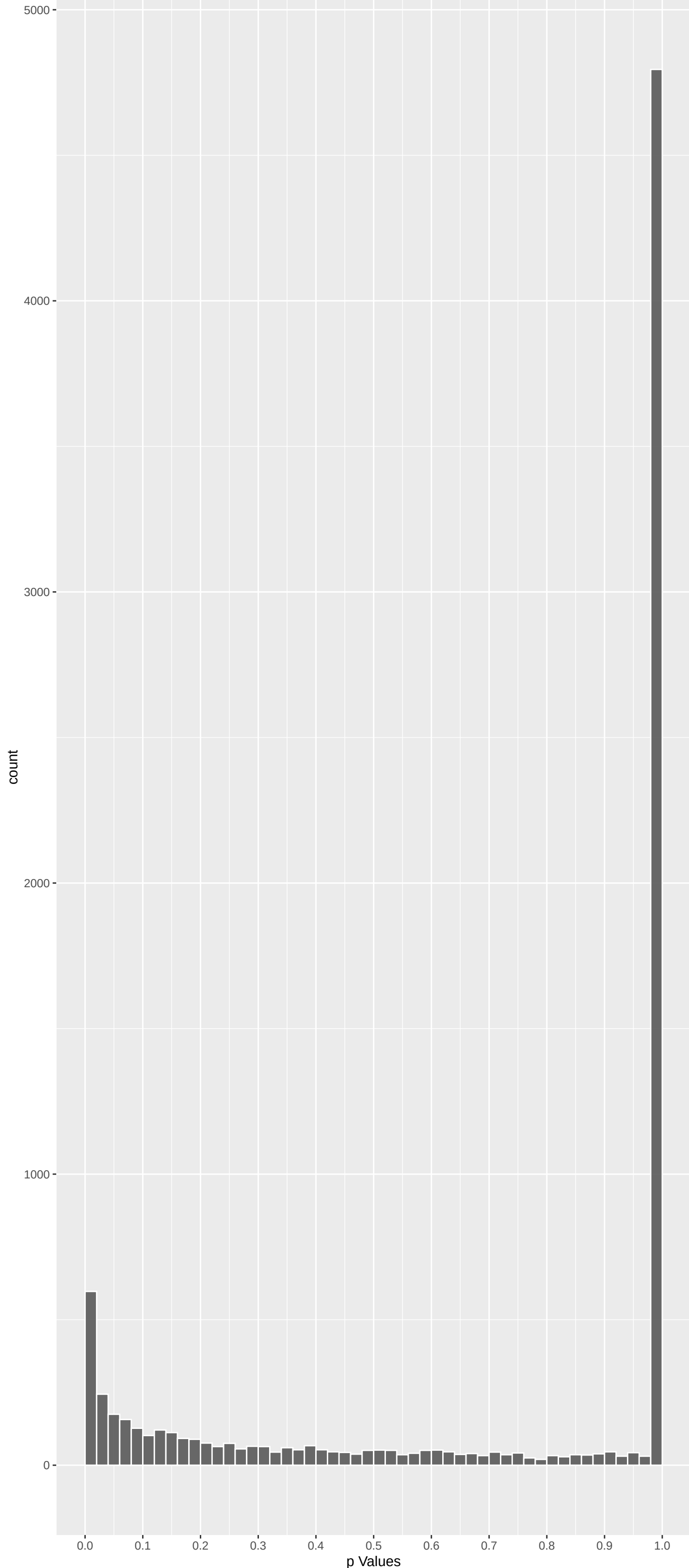
Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.559790	3.233009e-10	4.117e-06	4.214e-02
CPB1	6.383347	1.039551e-09	6.619e-06	4.214e-02
PGC	5.915927	1.980064e-08	8.405e-05	2.951e-01
TRPM6	5.852139	2.911744e-08	9.270e-05	2.951e-01
SLC5A12	5.381891	4.422455e-07	1.126e-03	1.000e+00
PTGFR	5.195582	1.224484e-06	2.599e-03	1.000e+00
DHDH	5.075971	2.313133e-06	2.678e-03	1.000e+00
NAGLU	5.085044	2.205245e-06	2.678e-03	1.000e+00
ADGRF1	5.093347	2.110782e-06	2.678e-03	1.000e+00
TET1	5.148424	1.576108e-06	2.678e-03	1.000e+00
INPP5K	5.119676	1.836364e-06	2.678e-03	1.000e+00
SERPINA7	4.901298	5.712319e-06	5.595e-03	1.000e+00
CTRC	4.916399	5.289039e-06	5.595e-03	1.000e+00
ADAM28	4.842789	7.681770e-06	6.987e-03	1.000e+00
MUC20	4.780381	1.049778e-05	8.355e-03	1.000e+00
CFAP20DC	4.785343	1.024174e-05	8.355e-03	1.000e+00
PRR14L	4.759827	1.162553e-05	8.708e-03	1.000e+00
MAN2A1	4.575040	2.856780e-05	1.915e-02	1.000e+00
JAKMIP1	-4.582319	2.759087e-05	1.915e-02	1.000e+00
ALB	4.552261	3.184345e-05	1.931e-02	1.000e+00
TULP4	-4.555768	3.131665e-05	1.931e-02	1.000e+00
CUBN	4.520981	3.693222e-05	2.138e-02	1.000e+00
ADCY1	-4.491900	4.235438e-05	2.247e-02	1.000e+00
CYP2R1	4.493380	4.206087e-05	2.247e-02	1.000e+00
CELA1	4.478495	4.510265e-05	2.297e-02	1.000e+00

Top genes by Q-Value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.559790363	3.233009e-10	4.117e-06	4.214e-02
CPB1	6.383347123	1.039551e-09	6.619e-06	4.214e-02
TRPM6	5.852139020	2.911744e-08	9.270e-05	2.951e-01
PGC	5.915927441	1.980064e-08	8.405e-05	2.951e-01
M6BP	0.003695055	1.000000e+00	1.000e+00	1.000e+00
FKBP4	0.546850545	1.000000e+00	1.000e+00	1.000e+00
NDUFAF7	1.897237101	3.467799e-01	1.000e+00	1.000e+00
FUCA2	1.043002907	1.000000e+00	1.000e+00	1.000e+00
DBNDD1	-2.007309140	2.683005e-01	1.000e+00	1.000e+00
HS3ST1	-0.048484763	1.000000e+00	1.000e+00	1.000e+00
SEMA3F	2.491535874	7.631527e-02	7.129e-01	1.000e+00
CFTR	1.477455448	8.373216e-01	1.000e+00	1.000e+00
CYP51A1	-0.035280503	1.000000e+00	1.000e+00	1.000e+00
USP28	-0.847404787	1.000000e+00	1.000e+00	1.000e+00
SLC7A2	0.105614640	1.000000e+00	1.000e+00	1.000e+00
PDK4	-1.661545680	5.796234e-01	1.000e+00	1.000e+00
USH1C	-0.294944468	1.000000e+00	1.000e+00	1.000e+00
RALA	-0.015058465	1.000000e+00	1.000e+00	1.000e+00
BAIAP2L1	0.333001100	1.000000e+00	1.000e+00	1.000e+00
CACNG3	-2.679640187	4.422080e-02	5.435e-01	1.000e+00
TMEM132A	-0.959277377	1.000000e+00	1.000e+00	1.000e+00
DVL2	-0.075907642	1.000000e+00	1.000e+00	1.000e+00
RPAP3	0.846511540	1.000000e+00	1.000e+00	1.000e+00
TRAPPC6A	0.094254084	1.000000e+00	1.000e+00	1.000e+00
SPATA20	1.388992929	9.890094e-01	1.000e+00	1.000e+00

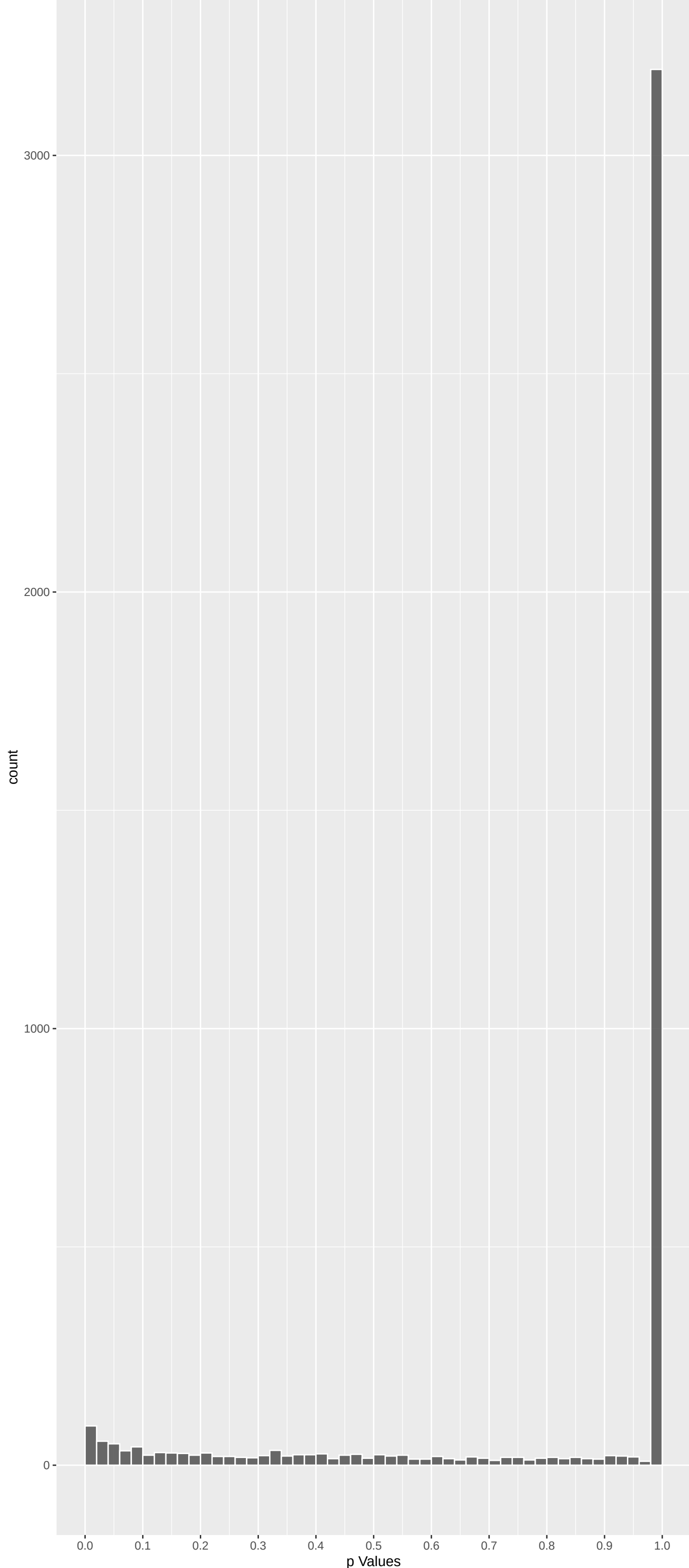
Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
BPIFB1	6.559790	3.233009e-10	4.117e-06	4.214e-02
CPB1	6.383347	1.039551e-09	6.619e-06	4.214e-02
PGC	5.915927	1.980064e-08	8.405e-05	2.951e-01
TRPM6	5.852139	2.911744e-08	9.270e-05	2.951e-01
SLC5A12	5.381891	4.422455e-07	1.126e-03	1.000e+00
PTGFR	5.195582	1.224484e-06	2.599e-03	1.000e+00
DHDH	5.075971	2.313133e-06	2.678e-03	1.000e+00
NAGLU	5.085044	2.205245e-06	2.678e-03	1.000e+00
ADGRF1	5.093347	2.110782e-06	2.678e-03	1.000e+00
TET1	5.148424	1.576108e-06	2.678e-03	1.000e+00

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
JAKMIP1	-4.582319	2.759087e-05	1.915e-02	1.000e+00
TULP4	-4.555768	3.131665e-05	1.931e-02	1.000e+00
ADCY1	-4.491900	4.235438e-05	2.247e-02	1.000e+00
CAMK2B	-4.344557	8.373435e-05	2.458e-02	1.000e+00
MYH13	-4.405850	6.322183e-05	2.458e-02	1.000e+00
TRPM5	-4.425900	5.762461e-05	2.458e-02	1.000e+00
EP400	-4.200314	1.599269e-04	3.211e-02	1.000e+00
VAX2	-4.120366	2.269630e-04	3.753e-02	1.000e+00
PLEKHF1	-4.083903	2.657128e-04	4.077e-02	1.000e+00
SORCS1	-3.993387	3.908168e-04	4.969e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
IRNA Methylation (GO:0030488)	0.25364110	34	3.150e-07	1.692e-03	TARBP1:75 TRMT9B:459 TRMT1L:460 LCMT2:474 TRMT10A:503 TRMT5:538
Fatty Acid Catabolic Process (GO:0009062)	0.18065022	55	5.695e-06	9.922e-03	EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 EC12:180 HDHA:245
Chemical Synaptic Transmission (GO:00072)	-0.09366238	188	1.009e-05	1.352e-02	GABRR1:16 STXB1:47 GABRB3:122 RIMBP2:128 RPS6KA2:138 CHRFA7A:139
Regulation Of Synaptic Transmission, Glu	-0.20396950	40	8.215e-06	1.353e-02	DGKI:24 PTK2B:63 MAPK8IP2:109 SHANK3:146 CACNG3:159 SYT1:199
Alpha-Amino Acid Catabolic Process (GO:1	0.22897878	29	1.999e-05	1.789e-02	ALDH4A1:80 TAT:187 HIBC:219 KMO:432 GORT2:558 ACMSD:791
Mitochondrial Gene Expression (GO:014005	0.13059664	91	1.743e-05	1.789e-02	PTCD3:67 TEFM:95 MRPS9:243 MRPL43:445 MRPS18A:504 MRPS22:599
NADH Dehydrogenase Complex Assembly (GO:	0.18870308	41	2.957e-05	1.925e-02	ECISIT:79 FOXRED1:98 NDUFAB:169 NDUBF8:308 DMAC2:347 TMEM126B:499
RNA Methylation (GO:0001510)	0.17311883	48	4.105e-05	1.925e-02	TARBP1:75 TRMT9B:459 LCMT2:474 RBM15B:480 TRMT10A:503 TFB2M:604
Fatty Acid Beta-Oxidation (GO:0006635)	0.17779355	45	3.757e-05	1.925e-02	EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 ACOX2:174 EC12:180
Mitochondrial Respiratory Chain Complex	0.18870308	41	2.957e-05	1.925e-02	ECISIT:79 FOXRED1:98 NDUFAB:169 NDUF88:308 DMAC2:347 TMEM126B:499
Non-Motile Cilium Assembly (GO:190515)	0.22186584	29	3.587e-05	1.925e-02	CEP89:210 ARL13B:282 CEP250:666 GORAB:887 ENKDI:903 C2CD3:1040
Steroid Metabolic Process (GO:0008202)	0.15246673	56	4.301e-05	1.925e-02	CYP2R1:21 CYP1A1:72 CYP26C1:138 DHRS4:194 CYP17A1:225 NPC1:277
Organic Hydroxy Compound Transport (GO:0	0.19316430	36	6.141e-05	2.537e-02	ABCG5:141 SLC51A:211 SLC51B:334 RALBP1:404 STRA6:466 SLC01A2:549
Cilium Organization (GO:0044782)	0.08572650	183	6.897e-05	2.646e-02	TCTN1:34 CDC66:71 CFPAP43:116 TCTN2:182 CEP89:210 IQUB:251
Mitochondrial Translation (GO:0032543)	0.12107935	87	9.829e-05	3.519e-02	PTCD3:67 MRPS9:243 MTIF2:269 MRPL43:445 GFMI:452 MRPS18A:504
Monocarboxylic Acid Transport (GO:001571	0.15778679	50	1.157e-04	3.656e-02	SLC5A12:5 SLC16A5:183 SLC22A13:205 SLC51A:211 SLC51B:334 SLC6A12:341
tRNA Modification (GO:0006400)	0.14344262	61	1.094e-04	3.656e-02	TARBP1:75 TRMT9B:459 LCMT2:474 TRMT10A:503 TRMT1:643 NSUN6:657
Mitochondrial Respiratory Chain Complex	0.13923641	62	1.532e-04	4.167e-02	ECISIT:79 FOXRED1:98 NDUFAB:169 NDUF88:308 DMAC2:347 TMEM126B:499
Modulation Of Chemical Synaptic Transmis	-0.11663539	89	1.480e-04	4.167e-02	DGKI:24 TRIO:77 MAPK8IP2:109 PTEN:151 GRDI:1172 BAIAP3:176
Positive Regulation Of Synaptic Transmis	-0.15330835	51	1.552e-04	4.167e-02	PTK2B:63 SHANK3:146 CACNG3:159 BAIAP3:176 CRHR2:186 SHANK2:188
Protein Glycosylation (GO:0006486)	0.10449613	107	1.961e-04	5.016e-02	TE11:7 B3GNT4:247 GALNT5:250 POGUT1:257 NPC1:277 FKTN:291
Double-Strand Break Repair (GO:0006302)	0.09222659	134	2.397e-04	5.852e-02	ANKLE1:168 WRN:207 ERCC5:259 NPC1:277 RNF168:526 PRKDC:671
Cell Junction Assembly (GO:0034329)	-0.13319888	63	2.617e-04	6.112e-02	SDK1:80 TLN1:94 SHANK3:146 PTEN:151 SHANK2:188 DBNL:295
Cilium Assembly (GO:0060271)	0.07535552	197	2.867e-04	6.333e-02	TCTN1:34 CDC66:71 CFPAP43:116 TCTN2:182 CDC339:197 CEP89:210
Nervous System Development (GO:0007399)	-0.06119661	301	2.948e-04	6.333e-02	VAX2:8 ADGRB1:27 SH3GL3:43 SCIN:60 CNTN4:70 SDK1:80
Monocarboxylic Acid Catabolic Process (G	0.24581269	18	3.071e-04	6.344e-02	CYP26C1:138 AGXT:626 AGXT2:1549 HOGAI:1721 IDUA:1865 LPLN1:1890
Muscle Contraction (GO:0006936)	-0.12129359	72	3.819e-04	7.598e-02	MYH13:5 TACR2:20 MYH2:116 LMOD1:267 MYH8:292 EMD:399
Anterograde Trans-Synaptic Signaling (GO)	-0.08739602	134	4.616e-04	8.717e-02	GABRR1:16 GABRB3:122 RPS6KA2:138 SYT1:199 KCNQ3:245 HRH3:526
Signal Release From Synapse (GO:0099643)	-0.18773908	29	4.707e-04	8.717e-02	STXB1:47 SYT1:199 SLC32A1:218 HRH3:326 CPLX1:346 NRXN2:341
Fatty Acid Oxidation (GO:0013995)	0.15246630	43	5.489e-04	9.828e-02	EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 EC12:180 HDHA:245
Bile Acid And Bile Salt Transport (GO:00	0.24608941	16	6.568e-04	1.080e-01	SLC51A:211 SLC16A:334 SLC01A2:549 SLC01A2:549 ATP8B1:1063 SLC01B1:1345
Central Nervous System Development (GO:0	-0.06807124	213	6.638e-04	1.080e-01	VAX2:8 SH3GL3:43 SCIN:60 CNTN4:70 PTEN:151 SHROOM2:168
Double-Strand Break Repair Via Homologou	0.10413494	90	6.581e-04	1.080e-01	FANCM:81 WRN:207 ERCC5:259 NPC1:277 RNF169:712 RAD54B:807
Glutamate Receptor Signaling Pathway (GO)	-0.19194577	26	7.097e-04	1.100e-01	PTK2B:63 HOMER2:325 TRPM1:356 GRM5:467 GRM6:538 GRIN2A:593
Recombinational Repair (GO:0000725)	0.11246635	76	7.165e-04	1.100e-01	WRN:207 ERCC5:259 NPC1:277 RHOH1:624 RNF169:712 RAD54B:807
Fatty Acid Metabolic Process (GO:0006631	0.09065472	103	7.817e-04	1.166e-01	ABCB11:65 CYP1A1:72 ACADM:92 CPT1C:97 PLA2G1B:146 ACOX2:174
Negative Regulation Of Fatty Acid Biosyn	0.36494792	7	8.274e-04	1.201e-01	TRIB3:315 ACADVL:362 BCRA1:932 UBRA4:2343 DCAF5:2724 SIRT1:2866
Fat-Soluble Vitamin Metabolic Process (G	0.23922218	16	9.266e-04	1.214e-01	CYP2R1:21 CYP1A1:72 CYP11A1:443 GGCKX:571 CYP2A41:655 LRAT:802
Nitrogen Compound Transport (GO:0071705)	0.08351818	133	9.200e-04	1.214e-01	CUBN:20 RHAG:185 SLC22A13:205 AQP9:228 RHCE:231 SLC6A6:256
Positive Regulation Of Synaptic Transmis	-0.26098919	13	8.963e-04	1.214e-01	PTK2B:63 SHANK3:146 CACNG3:159 GRIN2A:593 NPS:626 TSHZ3:724

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_DEVELOPMENTAL_BIOLOGY	-0.09071584	680	1.627e-15	1.053e-11	COL6A2:22 KRT20:37 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	-0.09599440	390	1.027e-10	3.326e-07	COL6A2:22 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78 TLN1:94
REACTOME_INFECTIOUS_DISEASE	-0.07237993	556	7.552e-09	1.630e-05	ADCY1:3 GOLGA7:26 DOCK1:38 SH3GL3:43 DYNC1H1:53 ITPR1:54
HOUNKPE_HOUSEKEEPING_GENES	-0.06326255	712	1.368e-08	1.772e-05	PHRF1:32 NSA2:36 HCF1:66 SND1:113 U2AF2:117 PTEN:151
REACTOME_NEURONAL_SYSTEM	-0.10094367	270	1.325e-08	1.772e-05	ADCY1:3 CAMK2B:6 GABRR1:16 STXB1:47 NBEA:51 AP2A2:78
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	-0.07536497	469	2.924e-08	3.156e-05	CAMK2B:6 EP400:7 DNAJA4:18 RPTOR:25 DYNC1H1:53 SRPRA:84
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	-0.08094527	378	7.967e-08	6.585e-05	COL6A2:22 DOCK1:38 SH3GL3:43 COL6A1:45 ITPR1:54 PTK2B:63
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	-0.07327046	464	8.135e-08	6.585e-05	CAMK2B:6 DOCKA:35 TCFL5:42 STXB1:47 DYNC1H1:53 TMED2:71
REACTOME_SIGNALING_BY_WNT	-0.11686042	161	3.344e-07	2.349e-04	ITPR1:54 PDE6B:74 AP2A2:78 PSMA3:205 TNRC6A:221 AGO3:280
WP_MRNA_PROCESSING	-0.18062307	66	3.999e-07	2.349e-04	U2AF2:117 SNRPB:156 SNRPA:161 PTBP2:163 SF3A3:270 SFSWAP:301
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.05465004	766	3.869e-07	2.349e-04	STXB1:47 DYNC1H1:53 ITPR1:54 PTPRE:61 PTK2B:63 TMED2:71
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	-0.11198787	170	5.083e-07	2.743e-04	SLBP:73 U2AF2:117 SNRPN:156 SNRPA:161 SF3A3:270 SFSWAP:301
BENPORATH_ES_WITH_H3K27ME3	-0.05465478	733	6.619e-07	3.297e-04	CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 SMPD3:23 DGKI:24
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	-0.08108163	297	1.734e-06	8.023e-04	CAMK2B:6 TLN1:94 SND1:113 HDAC4:118 PTEN:151 PSMA3:205
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	-0.12668765	118	2.092e-06	9.031e-04	ADCY1:3 CAMK2B:6 GABRR1:16 NBEA:51 AP2A2:78 GABRB3:122
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.05013951	785	2.477e-06	1.003e-03	PTGFR:6 ACY1:35 IDO2:51 ABCB11:65 CYP1A1:72 HKCES:77
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	-0.36237930	14	2.677e-06	1.020e-03	EP400:7 POLE:102 MMP17:237 GALNT9:261 SFSWAP:301 ANKLE2:394
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.21621579	38	4.029e-06	1.450e-03	ANK1:36 IDO2:51 RAB11FIP1:53 NSD3:64 DDHD2:111 IKBKB:135
REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC	0.25463672	26	7.026e-06	2.275e-03	CYP2R1:21 ACY1:35 CYP26C1:138 CYP17A1:225 CYP11A1:443 CYP2A41:655
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	-0.05052726	699	6.945e-06	2.275e-03	CAMK2B:6 IGFBP3:13 RPTOR:25 SLBP:73 ZNF704:86 USP2:89
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	-0.13948118	86	8.004e-06	2.468e-03	KLHL30:21 KIF1A:34 GYS1:68 HDAC4:118 TRAF3P1:131 GPC1:177
KIM_ALL_DISORDERS_CALB1_CORR_UP	-0.07010637	337	1.097e-05	3.088e-03	CAMK2B:6 DOCKA:35 STXB1:47 DYNC1H1:53 ITPR1:54 PTK2B:63
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.19407544	43	1.079e-05	3.088e-03	EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 ACOT12:109
REACTOME_FATTY_ACID_METABOLISM	0.10783639	139	1.191e-05	3.213e-03	CYP1A1:72 EHHADH:78 ACA2A:83 ACADM:92 ACAD11:106 ACOT12:109
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	-0.12650118	100	1.275e-05	3.304e-03	CABLES2:31 TCFL5:42 ZBTB46:201 RAE1:312 HRH3:326 MYT1:327
LIM_MAMMARY_STEM_CELL_UP	-0.06866132	364	1.340e-05	3.311e-03	SORCS1:10 UPP1:12 IGFBP3:13 PTPRE:61 TRIM29:81 RNF165:85
REACTOME_SIGNALING_BY_GPCR	-0.05637609	517	1.380e-05	3.311e-03	ADCY1:3 CAMK2B:6 TACR2:20 DGKI:24 GPR55:29 GAL40
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	-0.06088005	435	1.555e-05	3.597e-03	CAMK2B:6 ADGBR1:27 GAL40 CNTN4:70 MAPK8IP2:109 OTUD7A:114
BENPORATH_SUZ12_TARGETS	-0.04873091	684	1.774e-05	3.844e-03	VAX2:8 SORCS1:10 DGKI:24 HSF4:30 REPS2:41 CRABP1:57
BENPORATH_ED_TARGETS	-0.04902705	675	1.781e-05	3.844e-03	CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 DGKI:24 HSF4:30
REACTOME_INFLUENZA_INFECTION	-0.13931432	79	1.906e-05	3.982e-03	RAE1:312 RPL7A:408 RPS8:445 NUP93:463 NUP37:518 KPNAB3:564
REACTOME_BIOLOGICAL_OXIDATIONS	0.11162574	123	1.981e-05	4.008e-03	CYP2R1:21 ACY1:35 ALDH1A1:60 CYP1A1:72 CYP26C1:138 MGST3:215
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	-0.09439264	170	2.296e-05	4.506e-03	ADCY1:3 CAMK2B:6 GABRR1:16 STXB1:47 NBEA:51 AP2A2:78
WP_MALIGNANT_PLEURAL_MESOTHELIOMA	-0.07047266	301	2.869e-05	5.465e-03	RPTOR:25 TRAF2:55 HCF1:66 PTEN:151 PTK2:171 FZD10:198
LEIN_NEURON_MARKERS	-0.17006496	47	5.554e-05	1.028e-02	JAKMIP1:1 CAMK2B:6 REPS2:41 SNRPN:156 SYT1:199 DPP6:200
REACTOME_HIV_INFECTION	-0.09758035	142	6.206e-05	1.032e-02	AP2A2:78 PSMA3:205 RAE1:312 NUP93:463 FURIN:516 NUP37:518
REACTOME_GENE_SILENCING_BY_RNA	-0.13944262	69	6.304e-05	1.032e-02	MOV10L1:115 TNRC6A:221 AGO3:280 RAE1:312 NUP93:463 NUP37:518
REACTOME_TRANSPORT_OF_MATURE_TRANSCRIPT_	-0.15351799	57	6.185e-05	1.032e-02	SLBP:73 U2AF2:117 SYMPK:283 RAE1:312 SLU7:380 NUP93:463
WONG_MITOCHONDRIA_GENE_MODULE	0.09467620	151	6.191e-05	1.032e-02	CHDH:32 CRYZ:70 NDUFAB:169 SLC16A5:183 ATP13A3:249 GCDH:286
KEGG_SPLICEOSOME	-0.15742537	54	6.374e-05	1.032e-02	U2AF2:117 SNRPA:161 SF3A3:270 SMNDC1:331 SRSF10:354 SLU7:380

DisGenET Top pathways by non-permutation

Mitochondrial Diseases	0.07320435	308	1.173e-05	1.072e-01	ALB:19 TXNRD1:48 CRYZ:70 ECISIT:79 FOXRED1:98 NDUFAB:169
Schizophrenia	-0.03664206	1278	2.194e-05	1.072e-01	ADCY1:3 CAMK2B:6 GABRR1:16 DOCKA:35 TMEM132D:39 STXB1:47
Central neuroblastoma	-0.03773073	1086	4.686e-05	1.140e-01	CAMK2B:6 IGFBP3:13 KIF1A:34 DOCKA:35 GAL:40 ITPR1:54
Neuroblastoma	-0.03760430	1113	4.095e-05	1.140e-01	CAMK2B:6 IGFBP3:13 KIF1A:34 DOCKA:35 GAL:40 ITPR1:54
HIV Infections	-0.05230156	515	6.165e-05	1.204e-01	KRT20:37 TMEM132D:39 PTK2B:63 TMED2:71 VIPR1:103 PLEC:142
Cholestasis	0.07462788	229	1.089e-04	1.532e-01	ABCB11:65 SLC23A1:68 ABCG5:141 PLA2G1B:146 SLC51A:211 CYP17A1:225
Neurodevelopmental Disorders	-0.09339233	146	1.041e-04	1.532e-01	STXB1:47 ITPR1:54 HCF1:66 CNTN4:70 TRIO:77 GABRB3:122
Renal salt wasting	0.29337535	14	1.448e-04	1.768e-01	KCNJ1:363 SCN11A:442 CYP11A1:443 SARS2:547 SCN11B:756 CYP2A2:1154
Anemia, hereditary spherocytic hemolytic	0.46520717	5	3.153e-04	2.093e-01	ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA
Ciliopathies	0.08562931	150	3.110e-04	2.093e-01	TCTN1:34 TCTN2:182 CDC39:197 NHPH3:252 NME8:275 ARL1B:382
Downturned corners of mouth	-0.13956583	57	2.729e-04	2.093e-01	HERC2:111 HDAC4:118 SNRPN:156 SMCR1:396 CPLX1:346 SMCA1:383
Hyponatremia	0.18387534	32	3.214e-04	2.093e-01	CLDN16:66 SCN11A:442 CYP11A1:443 COMT:449 SARS2:547 MC2R:738
Liver carcinoma	-0.02566223	2215	2.144e-04	2.093e-01	EP400:7 IGFBP3:13 SMPD3:23 GAL:40 TCFL5:42 BNC2:50
Nephrocalcinosis	0.15779439	45	2.537e-04	2.093e-01	TRPM6:4 CYP2R1:21 CLDN16:66 SLC4A1:221 KCNJ1:363 GRHPR:385
Spherocytosis	0.46520717	5	3.153e-04	2.093e-01	ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA
Night blindness, congenital stationary	-0.21906490	21	5.131e-04	2.785e-01	PDE6B:74 USP11:91 GPR34:345 TRPM1:356 CABP4:454 GRM6:538
Autism Spectrum Disorders	-0.05360059	369	4.625e-04	2.785e-01	JAKMIP1:1 NBEA:51 ITPR1:54 CNTN4:70 TRIO:77 HERC2:111
Congenital cerebral hernia	0.16339271	38	4.969e-04	2.785e-01	TCTN2:182 FKTN:291 TMEM67:446 WDPCP:515 ARHGAP31:779 LRAT:802
Type 1 muscle fiber predominance	-0.36933337	7	7.152e-04	3.677e-01	COL6A2:22 COL6A1:45 NEB:59 TNNI1:406 MYH7:70 COL6A3:5124.5
Primary hyperoxaluria, type I	0.27900575	12	8.200e-04	4.005e-01	GRHPR:385 AGXT:626 DAO:1605 AGT:1648 PTHLH:1692 PRDX5:1771
Neoplasm Metastasis	-0.02211344	2446	9.539e-04	4.235e-01	UPP1:12 IGFBP3:13 COL6A2:22 SMPD3:23 ADGBR1:27 DOCKA:35
Undifferentiated carcinoma	-0.07649316	158	9.507e-04	4.235e-01	PTEN:151 SYT1:199 IL18:258 MYC:324 NSMF:365 KRT71:366
Cholestasis in newborn	0.10520772	80	1.167e-03	4.768e-01	ABCB11:65 EHHADH:78 ABCG5:141 HDHA:245 NHPH3:252 NPC1:277
Fragile skin	-0.20972631	20	1.172e-03	4.768e-01	PLEC:142 ITGB4:388 PKP1:465 DSP:638 COL5A1:669 ITGA3:741
Aneurysm, Ruptured	0.34825540	7	1.420e-03	4.911e-01	RNH1:106 PPLR1:3L:504 SHC3:514 NOS3:389 ACE:975 FN1:5124.5
Neuritis, full face	-0.12297485	56	1.478e-03	4.911e-01	COL6A2:22 COL6A1:45 HDAC4:118 SEMA5:133 PTEN:151 SH3BP2:251
Anaplastic Ependymoma	-0.22918710	16	1.508e-03	4.911e-01	IGFBP3:13 PTEN:151 SMARCB1:621 NOTCH1:672 TIMP3:921 FOXJ1:998
Glioma	-0.02666157	1392	1.363e-03	4.911e-01	CAMK2B:6 IGFBP3:13 RPTOR:25 ADGBR1:27 DOCK1:38 GAL:40
Polycystic	0.18146915	26	1.369e-03	4.911e-01	CLDN16:66 NHPH3:252 KCNJ1:363 TMEM67:446 SLC12A3:546 SARS2:547
Round face	-0.12297485	56	1.478e-03	4.911e-01	COL6A2:22 COL6A1:45 HDAC4:118 SEMA5:133 PTEN:151 SH3BP2:251
Hyperkalemia	0.23530738	15	1.608e-03	5.066e-01	CYP17A1:225 SCN11A:442 CYP11A1:443 INVS:697 SCN11B:756 WNK4:1128
Metabolic acidosis	0.11558389	62	1.679e-03	5.126e-01	EHHADH:78 ACADM:92 LCT:133 SLC4A1:221 GCDH:286 SLC9A4:403
EMG: neuropathic changes	-0.23105860	15	1.951e-03	5.717e-01	DYHNC1:53 NEB:59 HSP33:405 VRY1:530 CAPH:352 MYH7:70
Glioblastoma	-0.02683034	1258	2.024e-03	5.717e-01	IGFBP3:13 DGKI:24 ADGBR1:27 DOCK1:38 GAL:40 TCFL5:42
Lethargy	0.09449603	90	1.991e-03	5.717e-01	ACADM:92 FOXRED1:98 SLC4A1:221 ACADSB:293 CPT2:314 ACADVL:362
Mammary Neoplasms	-0.02412423	1578	2.010e-03	5.717e-01	CAMK2B:6 IGFBP3:13 SMPD3:23 KRT20:37 DOCK1:38 GAL:40
Purpura	0.14454806	38	2.054e-03	5.717e-01	ITGA2B:409 GP1BB:476 GP9:825 MEV:939 NFkB2:1013 PRKCD:1018
Severe myopia	-0.10263103	77	1.888e-03	5.717e-01	IGFBP3:13 PDE6B:74 OPN1LW:236 TRPM1:356 VIPR2:423 CABP4:454
Sprains and Strains	0.09539962	5	1.975e-03	5.717e-01	PTEN:151 IL18:258 XIAP:268 TMEM1:441 TRO:5124.5 NA
Pain, Postoperative	0.20903077	18	2.146e-03	5.218e-01	COMT:449 TAOK3:701 APB1:1566 TNFRSF1B:1852 CPOX:1981 PTGS2:1988